NACS 645 –
The neural need to infer others

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Two-player strategic games

Stag Hunt

	Stag	*Hare*
Stag	8,8	0,7
Hare	7,0	5,5

1 Nash equilibrium

Side of the road

	Left	*Right*
Left	1,1	0,0
Right	0,0	1,1

2 Nash equilibria

Pure coordination

Prisoners' dilemma

	C	*D*
C	-1,-1	-4,0
D	0,-4	-3,-3

1 Nash equilibrium

Battle of the sexes

	B	*F*
B	2,1	0,0
F	0,0	1,2

2 Nash equilibria

Mixed motive: coop / competition

Matching pennies

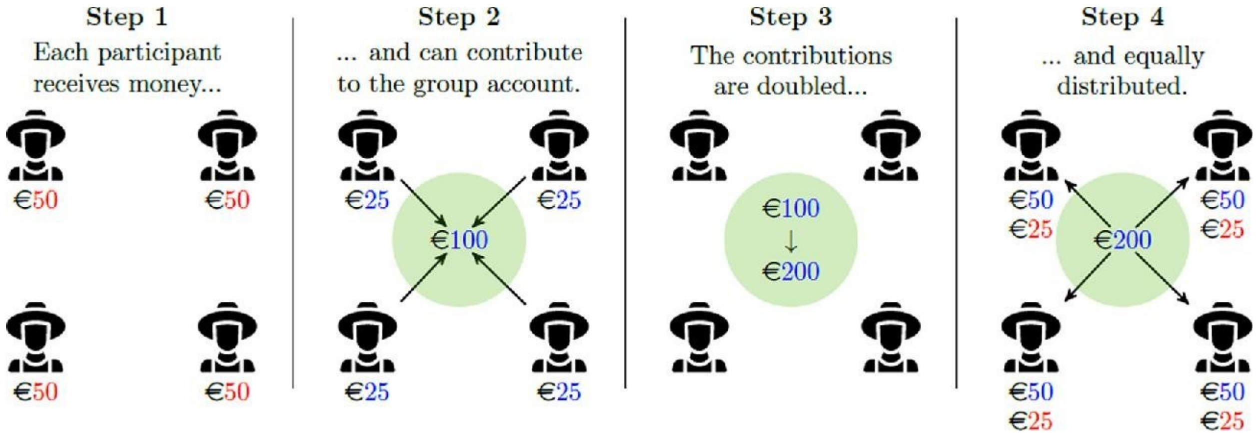
	Heads	*Tails*
Heads	1,-1	-1,1
Tails	-1,1	1,-1

0 Nash equilibrium

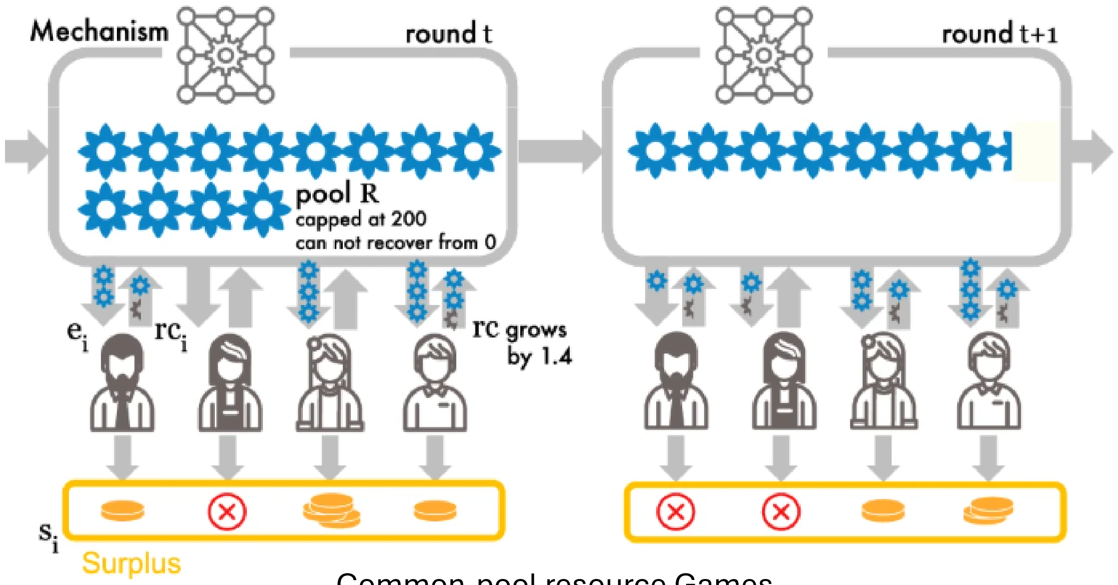
Pure competition

- In **strategic environments with mixed or competitive motives, deterministic strategies are exploitable**: predictability creates information asymmetry that can be exploited
- **Cooperation emerges** when it is **incentivized**, and is supported by **direct reciprocity**: **Mutual monitoring** and **repeated interactions** allow conditional cooperation (Best Responses are identifiable)
- Expectations and emotions may act as proximate regulators

N-player strategic games



Public Goods Game.
Adapted from Rommel et al., 2022. *Q Open*.



Common-pool resource Games.
Adapted from Koster et al., 2025. *Nature Communications*.

- In n-player strategic games with mixed motives, not only **deterministic strategies are exploitable**, but **games are hardly tractable**, and **often incomplete** (Bayesian)
  There is a high incentive for free-riding, which is responsible for cooperation collapse.
- **Cooperation emerges** when it is **incentivized**, and is **supported by indirect reciprocity**.
  Cooperation is sustained when **shared norms** appear and when **altruistic punishment** is allowed (external regulators); otherwise free-riders might trigger cascades of defection.

Cooperation at scale

Societal coordination

- In large, anonymous populations, direct and indirect reciprocity are **insufficient**
- Cooperation depends on **shared norms, social identity, and institutional enforcement**

Norms and expectations

- Norms are **shared rules sustained by mutual expectations and enforcement**. Abiding requires:
  - (i) **one believes most others follow the rule**
  - (ii) **one believes most others *approve* of the rule and will sanction non-compliance**
- Large-scale cooperation hinges on perceived compliance and enforcement
  Without, probability estimates of others' cooperation decrease, lowering the *subjective value* of cooperating

Institutions and identity

- **Social identity** extends mutual trust to symbolic communities
- **Institutions** (laws, contracts, markets, formal punishment systems) replace interpersonal reciprocity with credible enforcement. They act by:
  - **reshaping payoffs** (incentives for cooperation)
  - **altering expectations** (perceived probability that others will cooperate)
- Institutional trust built on norms and shared identity
  Erosion of legitimacy through unfairness, corruption, or unequal enforcement

Van Bavel et al., 2022. *Advances in experimental social psychology.*

Rand & Nowak, 2013. *TICS.*
Fehr & Schurtenberger, 2018. *Nature Human Behavior.*
Greene & Young, 2020. The Cog. Neuro. of Moral Judgment and Dec.-Making
Curry et al., 2022. *Review of Philosophy and Psychology*

Pathways to cooperation

Evolutionary Pathways for Cooperation
(selection mechanisms for cooperation; cf. Rand & Nowak, 2013)

- **Kin selection**
- **Direct reciprocity**
- **Indirect reciprocity**
- **Spatial selection**
- **Group selection**

Non-moral social emotions act as indirect or local regulators: manage reputation, kinship, coalitions, but are not moral obligations.

Cultural and Institutional Reinforcers
(external mechanisms for cooperation within societies)

- **Altruistic punishment** – Costly sanctioning of free riders
- **Social norms** – *Known standards of behavior based on shared beliefs about how individuals ought to behave in a specific context*
- **Rules and institutions** – Formal codifications of norms with enforcement mechanisms

Moral machinery
(evolved psychological and motivational mechanisms for internally regulated cooperation)

- Morality as a collection of biological & cultural solutions that promote and sustain cooperation.
- Moral cognition integrates valuation, reasoning, control, emotions to regulate cooperative behaviors.
- Moral emotions act as internalized regulators, by operating as *psychological carrots and sticks*. *e.g., Guilt* → self-directed punishment; *Gratitude* → reward toward others

Overview of morality-as-cooperation.

	Label	Problem/Opportunity	Solution
1	Family	Kin selection	Kin Altruism
2	Group	Coordination	Mutualism
3	Reciprocity	Social Dilemma	Reciprocal Altruism
4	Heroism	Conflict Resolution (Contest)	Hawkish Displays
5	Deference	Conflict Resolution (Contest)	Dove-ish Displays
6	Fairness	Conflict Resolution (Bargaining)	Division
7	Property	Conflict Resolution (Possession)	Ownership

Table 2 Twenty-one moral molecules

	Mutualism	Exchange	Hawk	Dove	Division	Possession
Kinship	Fraternity	Blood Revenge	Family Pride	Filial Piety	Gavelkind	Primo-geniture
Mutualism		Friendship	Patriotism	Tribute	Diplomacy	Common ownership
Exchange			Honour	Confession	Turn-taking	Restitution
Hawk				Modesty	Mercy	Munificence
Dove					Arbitration	Mendicance
Division						Queuing

M-a-C predits 127 moral & 127 immoral (up to seventh-order combination) with a periodic-table system, to solve as many cooperation problems.

We can rely on many external and internalized devices to set up expectations, but reasons to mentalize others don’t disappear.

Mutual monitoring, moral norms, institutions, and the mere individual willingness to follow rules don’t guarantee one will meet their expected utility.

In situation of mixed motives, of imperfect information (hidden actions, states or rewards), and/or incomplete (Bayesian) information, one needs to infer causality relationships between actions and intents.

This involves estimating actions, rewards, identities, personalities, preferences, mental states.

Mentalizing, central to strategic interactions

Tomasello, 2008. *MIT Press*.
Tomasello, 2020. *Episteme*.
Sperber et al., 2010. *Mind & Language*.

Cooperative communication

- Aligning beliefs and actions requires mutual **transparency** of minds
- **Language evolved for coordination and shared understanding** (Tomasello, 2008, 2020): sharing mental states, referencing *what is*, building ground truth in common reality
- **But transparency allows exploitation**: free riders gain if they remain undetected

Need for truthfulness

- Stable cooperation requires a) **most communication be truthful**; and b) **truth is the default expectation**
- Yet, this enables strategic deception and exploiting others
- Too many free riders → collapse of cooperation

Dual pressures

- Mentalizing makes cooperation possible and deception feasible
- Lying is effortful because it requires simulating others' minds
- **Communication evolved** under **dual pressures**: **cooperation** through truth-sharing **vs. competition** through manipulation

Epistemic vigilance

- **To protect cooperation, evolution also favored mechanisms for detecting dishonesty** (Sperber, 2010)
- Cognitive systems that assess reliability and sincerity of communicated information, acting as counterweights to gullibility

The development of ToM

Bettle & Rosati, 2021. *Language learning and Development.*
Fujita, Devine, Hughes, 2022. *Cognitive Development.*
Rakoczy, 2022. *Nature Reviews Psychology.*

(shared with other primates)

Simple gaze following →

Goal and intention attribution →

Level 1 Perspective taking (presence/absence) →

Attribute knowledge and ignorance →

(human-unique?)

Joint attention; sustained attention to others

Attributing cooperative or shared intentions

Level 2 perspective-taking ('how' it looks)

Attribute false beliefs

function

- Recognize positive intentions
- Share attention with others

- Identify cooperative opportunities
- Understand helpful communication
- Maintenance so all parties benefit

- 'Bird's-eye view' of joint acts
- Facilitate coordination

- Facilitate coordination
- Avoid miscommunication
- Reputation management

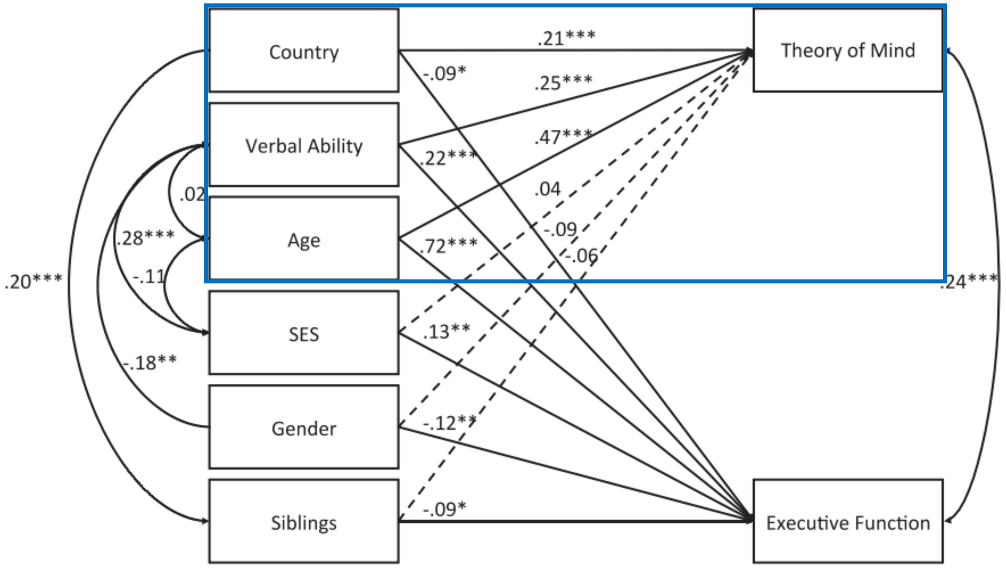


Fig. 1. A structural equation model of cross-cultural differences in theory of mind and executive function with covariates. Country: 0 = Japan, 1 = UK. Gender: 0 = girls, 1 = boys. SES = Socioeconomic Status. *Note.* * $p < .05$. ** $p < .01$. *** $p < .001$.

9 month revolution | **4 year revolution**

Birth → Age 9 months → Age 1–3 years → Age 4 years → Into adulthood →

Age 9 months:
- Basic ToM: perception–goal psychology
- Shared intentionality

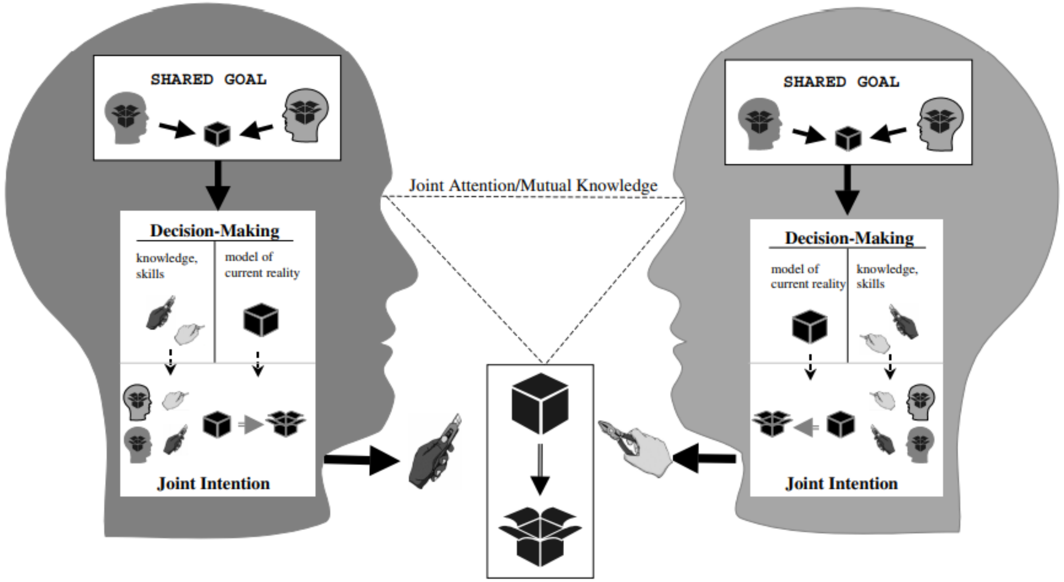
Age 1–3 years:
- Refinement of basic ToM

Age 4 years:
- fully fledged meta-representation ToM
- Belief–desire psychology

Into adulthood:
- Refinement of meta-representation
- Recursive, higher-order ToM

Understanding and sharing intentions

Tomasello et al., 2005. *Behavioral and brain sciences.*



Human's conception of a collaborative activity (shared goal & joint intention)

Ape general ontogenetic pathway of understanding

Understand others as: a) animate, b) goal-directed and c) intentional agents

Human-specific ontogenetic pathway of motivation

Motivation to share: a) emotions, b) experience and c) activities with others

Human sociality as **intersecting** both pathways: understanding intentional actions & motivation to share states. Intersection between 9mo and 14mo.

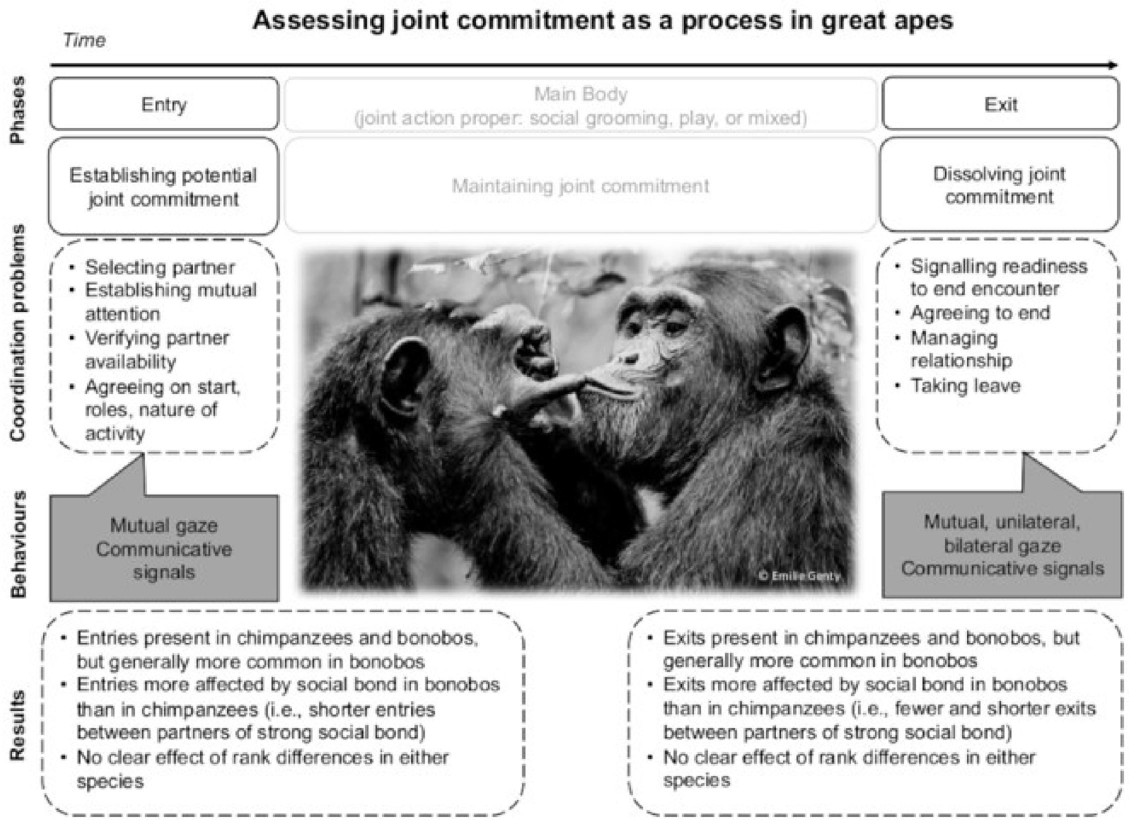
Tomasello et al. proposed that:

- The key distinction of human cognition is the capacity for **shared intentionality**: **participating in collaborative activities through mutually recognized goals and intentions**
- This participation **requires intention reading, cultural learning, and a motivation to share psychological states**
- **As a result, enables representations such as language, norms and collective beliefs**
- Primates grasp individual intentions. Humans integrate individual intentions into *joint commitments*

Heesen et al., 2021. *iScience*.

Joint commitment

Shared sense of obligation between co-actors in a common activity



- **Great apes show proto-commitment behaviors during joint actions** (e.g., play, grooming)
  They coordinate entry, maintenance, and exit phases, using gaze, gestures, and pauses to negotiate participation
- These process-level commitments suggest a **graded evolutionary continuity** with humans
  Bonobos show phases more moderated by friendship than chimpanzees ("face management" –like pattern)
- **Humans display normative, institutionalized forms of commitment** (conventions, roles, agreements) that allow stable, long-term collaboration and cultural accumulation

- Coordination capacities are not strictly uniquely human, but may explain the shift from coordination for mutual benefit to cooperation under shared obligation, and the capacities for complex mentalizing